JAVA PROGRAMMING LAB (23AI5351) SIDDHARTHA ACADEMY OF HIGHER EDUCATION (DEEMED TO BE UNIVERSITY) • DEPARTMENT OF AI • BATCH 2025–2029

Week 07: Multithreading & Thread Synchronization

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1. OBJECTIVE / AIM

To demonstrate Multithreading by extending Thread class, implementing Runnable interface, setting thread priorities, and utilizing synchronized methods.

2. CONCEPT & THEORY

Threads enable concurrent program execution to maximize multi-core processor utilization. Synchronization guarantees thread safety by serializing access to shared resources.

3. JAVA SOURCE IMPLEMENTATION

```
// Java Programming Lab - Experiment 07
// Student: Trivikram | Roll No: 25EU02067
class CounterTable {
    synchronized void printTable(int n) {
        for (int i = 1; i <= 3; i++) {
            System.out.println(Thread.currentThread().getName() + " -> " + n + " * "
            try {
                Thread.sleep(300);
            } catch (InterruptedException e) {
                System.out.println(e);
            }
        }
    }
}

public class Week7Demo {
    public static void main(String[] args) {
        CounterTable table = new CounterTable();

        Thread t1 = new Thread(() -> table.printTable(5), "Worker-1");
        Thread t2 = new Thread(() -> table.printTable(20), "Worker-2");

        t1.start();
        t2.start();
```

```
    }
}
```

4. COMPILATION & EXECUTION OUTPUT

```
Worker-1 -> 5 * 1 = 5
Worker-1 -> 5 * 2 = 10
Worker-1 -> 5 * 3 = 15
Worker-2 -> 20 * 1 = 20
Worker-2 -> 20 * 2 = 40
Worker-2 -> 20 * 3 = 60
```